

The empirical recurrence for OEIS A302068, recovered from its tail

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Abstract

OEIS A302068 records “Empirical recurrence of order 53” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 627$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 57$. Its last coefficient is $c_{53} = -8640 \neq 0$, so no further tail satisfies anything shorter; any order-53 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A302068 is “Number of $n \times 7$ 0..1 arrays with every element equal to 0, 1, 2 or 4 horizontally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

64, 2778, 89658, 3247765, 119508742, 4418931065, 163660519064, 6065045335103, ...

The entry states:

Empirical recurrence of order 53 (see link above)

As of the “Last modified” line on the live entry (Mar 31 2018 15:49:51, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 627$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 627$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 53: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 53.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1254$ exact terms from $n = 4$ on, it returns order 53 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{53} c_i a(n-i),$$

c_1	128	c_{15}	10629156707128106	c_{29}	5783133222214615422	c_{43}	−176447928573414
c_2	−7246	c_{16}	−32099440863467277	c_{30}	−4475298862336599878	c_{44}	46797282011633
c_3	247300	c_{17}	86674135815935428	c_{31}	3134104987685810856	c_{45}	−10538703960190
c_4	−5807860	c_{18}	−209994777042619011	c_{32}	−1973112653216358410	c_{46}	1999157966218
c_5	101188194	c_{19}	458039578388088288	c_{33}	1104957794101374358	c_{47}	−314969940878
c_6	−1373694658	c_{20}	−902262377910173841	c_{34}	−540554840665564592	c_{48}	40356202976
c_7	15042457178	c_{21}	1609718450314041488	c_{35}	222994776003762740	c_{49}	−4079581436
c_8	−136280615772	c_{22}	−2607782370371610391	c_{36}	−71054013746452527	c_{50}	311239920
c_9	1041076557756	c_{23}	3844593400915197640	c_{37}	11899085663268308	c_{51}	−16717888
c_{10}	−6803037312861	c_{24}	−5167120781669960190	c_{38}	4500809381677050	c_{52}	558528
c_{11}	38446730992396	c_{25}	6338828772254267768	c_{39}	−5620989209512984	c_{53}	−8640
c_{12}	−189515124191127	c_{26}	−7102790746989557919	c_{40}	3382418224338892		
c_{13}	820313847898546	c_{27}	7269891687445460666	c_{41}	−1528164967958718		
c_{14}	−3135134956137696	c_{28}	−6792083618385364099	c_{42}	565604829461721		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 57$, and it is the recurrence OEIS A302068 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{53} - c_1 x^{53-1} - \dots - c_{53}$. Its constant term is $-c_{53} = 8640$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 53 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 53, so $\deg q = 53$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 53, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 14 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 53, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -8640 .

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-53 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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